FLOODING AND SOCIETY: SURVEY INSTRUMENT
Date: June 12-18, 2024; Respondents = 1,200; Median Time = 19 min

You are invited to participate in a research study entitled "Severe Weather and Society." This study seeks to assess how U.S. residents receive, understand, and respond to weather forecasts and warnings. You were selected as a possible participant because you volunteered to participate in online surveys through Verasight or one of its partners. If you agree to participate, you will complete this approximately 20-minute online survey. There are no risks or benefits.

If you participate, you will be compensated according to your agreement with your online survey provider. Your participation is voluntary, and your responses will be de-identified before they are shared for research purposes or published.

Even if you choose to participate now, you may stop participating at any time and for any reason. Your data may be used in future research studies unless you contact me to withdraw your data.

Data are collected via an online survey system that has its own privacy and security policies for keeping your information confidential. The University of Oklahoma cannot provide assurances as to how this online survey system is permitted to use the data you provide.

If you have questions about this research, please contact the Institute for Public Policy Research and Analysis at the University of Oklahoma, at (405) 325-1720 or at ippra@ou.edu.

You can also contact the University of Oklahoma – Norman Campus Institutional Review Board at 405-325-8110 or irb@ou.edu with questions, concerns, or complaints about your rights as a research participant, or if you don't want to talk to the researcher.

By answering the survey questions, I agree to participate in this research. Please print this page for your records. This research has been approved by the University of Oklahoma, Norman Campus IRB.

IRB Number: 9418
Approval date: 10/18/2023

----------End Web pg----------

age: How old are you? [DROPDOWN; Less than 18 to 110; IF < 18 SKIP TO END OF SURVEY]

gend: Are you male or female?
0 - Female
1 - Male

hisp: Do you consider yourself to be Hispanic, Latino, or Spanish or to have Hispanic, Latino, or Spanish origins?
0 - No
1 - Yes

race: Which of the following best describes your race?
1 - White
2 - Black or African American
3 - American Indian or Alaska Native
4 - Asian
5 - Native Hawaiian or Pacific Islander
6 - Two or more races

----------End Web pg----------

state: Please select the state or district where your current residence is located.

zip: What is the five-digit zip code at your residence? [VERBATIM; REQUIRE 5-DIGIT NUMERIC]

----------End Web pg----------

Approximately how long have you lived at your current address or any other address within the same zip code area?
long_years: [DROPDOWN; Less than 1 year to 100]

----------End Web pg----------

[SHOW IF **long_years** < 5]

last_state: Using the dropdown list, please select the state or district where your previous residence was located.

----------End Web pg----------

[SHOW IF **last_state** = Other]

last_state_othr_spec: Other location (please specify) [VERBATIM]

----------End Web pg----------

[SHOW IF **last_state** =/ Other]

last_zip: What was the five-digit zip code where your previous residence was located? [VERBATIM; REQUIRE 5-DIGIT NUMERIC]

----------End Web pg----------

now: Please indicate which of the following statements applies to you.
0 - I am completing this survey from my current primary residence.
1 - I am completing this survey from a location that is not my current primary residence.

----------End Web pg----------

rural: Which of the following categories best describes the location of your current primary residence?
1 - Urban lot in a densely populated area
2 - Suburban lot in a neighborhood that is near a densely populated area
3 - Rural lot in a sparsely populated area

home: Which of the following categories best describes the nature of your current primary residence?
1 - Stand-alone (detached) permanent structure such as a house
2 - Condominium, townhouse, or duplex that is attached to another structure
3 - Apartment or dormitory room that is part of a larger residential complex
4 - Mobile home (whether placed on a permanent foundation or not)
5 - Boat, boathouse, ship, dock, or other floating structure
6 - Other type (please specify)
home_spec: [VERBATIM]

rent: Which of the following categories best describes your living arrangements at your current primary residence?
1 - Live with family or friends (e.g., in someone else's home) and do not pay rent
2 - Pay to rent or lease your primary residence (includes college or other dormitory rooms)
3 - Own your primary residence (includes making mortgage payments or outright ownership with no mortgage payments)

----------End Web pg----------

adults: *Including yourself,* how many *adults* age 18 and older live in your current primary residence? [DROPDOWN;1 to More than 20]

children: How many *children* age 17 and younger live in your current primary residence? [DROPDOWN; 0 to More than 20]

----------End Web pg----------

Now we have some basic questions about the weather. How much do you agree or disagree with the following statements? [RANDOM ORDER]

follow: I follow the weather very closely.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

plan_around: I plan my daily routine around the weather.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

und_weather: I don't understand what causes extreme weather events like blizzards, tornadoes, and hurricanes.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

----------End Web pg----------

How frequently do you get information about the weather from each of the following sources? [RANDOM ORDER IN TABLE]

wthr_info_paper: Newspapers
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_web: Non-government Internet websites (such as weather.com)
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_govweb: Government Internet websites (such as weather.gov)
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_loctv: Local TV (television) news
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_cabtv: Cable TV (television) news (such as The Weather Channel)
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_radio: Radio
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_fam: Family, friends or colleagues
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_soc: Social Media, such as Facebook
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_app: Weather applications on your phone
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

wthr_info_text: Emergency notifications on your phone
1 - Never
2 - Less than once per week
3 - About once per week
4 - Several times per week
5 - About once a day
6 - Several times a day

----------End Web pg----------

Thinking about all four seasons (spring, summer, fall, winter), how do you rate the risk of the following hazardous weather events to you and the people in the area where you live? [RANDOM ORDER IN TABLE; REQUIRE ALL]

risk_hail: Hail
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_wind: High winds
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_lignt: Lightning
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_heat: Heat waves
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_drought: Drought
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_cold: Cold temperatures
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_snow: Snow

1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_ice: Ice or freezing rain
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_tor: Tornadoes
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_flood: Floods
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_hur: Hurricanes
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_fire: Wildfires
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

risk_bliz: Blizzards
1 - No risk
2 - Low risk
3 - Moderate risk
4 - High risk
5 - Extreme risk

----------End Web pg----------

risk_tie: Please indicate which type of event poses the *biggest* risk to you and the people in the area where you live.
1 - Hail
2 - High winds
3 - Lightning
4 - Heat waves
5 - Drought

6 - Cold temperatures
7 - Snow
8 - Ice or freezing rain
9 - Tornados
10 - Floods
11 - Hurricanes
12 - Wildfires
13 - Blizzards

----------End Web pg----------

The National Weather Service (NWS), an agency of the United States government, issues weather forecasts and different kinds of alerts to the public about hazardous weather, including severe weather watches, warnings, and advisories.

alert_und: In general, do you understand the difference between watches, warnings, and advisories?
1 - Definitely no
2 - Probably no
3 - Not sure
4 - Probably yes
5 - Definitely yes

----------End Web pg----------

The next few questions focus on floods. For simplicity, we use the word flood to reference all kinds of flooding caused by weather conditions, including flash floods, riverine floods, coastal flooding, and urban flooding.

----------End Web pg----------

To begin, please think about you, family members you live with, and your home as you respond to the following questions about flooding.

fl_risk_prob: How likely do you think it is that the area where you live will be affected by a flood in the next few months?
1 - Not at all likely
2 - Not very likely
3 - Somewhat likely
4 - Very likely
5 - Extremely likely

fl_risk_imp: If the area where you live is affected by a flood in the next few months, how likely do you think it is that you personally will experience negative impacts from the flood?
1 - Not at all likely
2 - Not very likely
3 - Somewhat likely
4 - Very likely
5 - Extremely likely

fl_risk_sev: If the area where you live is affected by a flood in the next few months, how severe do you think the negative impacts will be to you from the flood?
1 - Not at all severe
2 - Not very severe
3 - Somewhat severe
4 - Very severe
5 - Extremely severe

fl_risk_aff: If the area where you live is affected by a flood in the next few months, how worried are you that it will affect you?
1 - Not at all worried
2 - Not very worried
3 - Somewhat worried
4 - Very worried
5 - Extremely worried

----------End Web pg----------

The National Weather Service issues multiple public alerts during floods depending on what type of flooding is occurring. We want to know how you interpret these different types of alerts.

flood_wwa_und: How would you rate your understanding of *flood* WATCHES and WARNINGS?
1 - Poor
2 - Fair
3 - Good
4 - Very good
5 - Excellent

flash_wwa_und: How would you rate your understanding of *flash flood* WATCHES and WARNINGS?
1 - Poor
2 - Fair
3 - Good
4 - Very good
5 - Excellent

----------End Web pg----------

To the best of your knowledge, is the following alert considered a flood WATCH or flood WARNING? [RANDOM SPLIT; 50% get **flood_wwa_watch**; 50% get **flood_wwa_warn**]

flood_wwa_watch: This alert is issued when conditions are favorable for flooding. It does not mean flooding will occur, but it is *possible*.
1 - Flood WATCH
2 - Flood WARNING
3 - Don't know

flood_wwa_warn: This alert is issued when flooding is imminent or occurring.
1 - Flood WATCH
2 - Flood WARNING
3 - Don't know

----------End Web pg----------

flood_warn_time: If the National Weather Service issues a *flood WARNING* for your location, about how much time do you have before flooding begins?
1 - less than 1 hour
2 - 1 to 24 hours
3 - 1 to 2 days
4 - 2 to 4 days
5 - 4 to 8 days

flood_watch_time: If the National Weather Service issues a *flood WATCH* for your location, about how much time do you have before flooding begins?
1 - less than 1 hour

2 - 1 to 24 hours
3 - 1 to 2 days
4 - 2 to 4 days
5 - 4 to 8 days

----------End Web pg----------

flood_flash_und: Do you understand the difference between *floods* and *flash floods*?
1 - Definitely no
2 - Probably no
3 - Not sure
4 - Probably yes
5 - Definitely yes

flood_flash_desc: Please use a few words to explain the primary difference between floods and flash floods. [VERBATIM]

----------End Web pg----------

To the best of your knowledge, is the following alert considered a FLOOD warning or a FLASH FLOOD warning? [RANDOM SPLIT; 50% get **flood_warn**; 50% get **flash_warn**]

flood_warn: This alert is issued for flooding that normally occurs more than six hours after the rainfall begins, and may cover a large area. This results in a gradual ponding or buildup of water in low-lying, flood prone areas, as well as small creeks and streams.
1 - FLOOD warning
2 - FLASH FLOOD warning
3 - Don't know

flash_warn: This alert is issued for flooding that normally occurs within six hours of heavy or intense rainfall. This results in small creeks and streams quickly rising out of their banks.
1 - FLOOD warning
2 - FLASH FLOOD warning
3 - Don't know

----------End Web pg----------

For people who live on the coast of an ocean, coastal flooding and storm surge are also a concern.

coast_live: Do you live on the coast of an ocean?
0 - No
1 - Yes

coast_risk_prob: How likely do you think it is that the area where you live will be affected by *coastal flooding* in the next few years?
1 - Not at all likely
2 - Not very likely
3 - Somewhat likely
4 - Very likely
5 - Extremely likely

surge_risk_prob: How likely do you think it is that the area where you live will be affected by *storm surge* in the next few years?
1 - Not at all likely
2 - Not very likely
3 - Somewhat likely

4 - Very likely
5 - Extremely likely

flood_srg_und: For some people, it is difficult to explain the difference between flooding and storm surge. How would you rate your understanding of the difference between the two hazards?
1 - Poor
2 - Fair
3 - Good
4 - Very good
5 - Excellent

----------End Web pg ----------

The next set of questions is about the variety of forecast information that you might receive when a flood threatens your location. If you have not been in this situation, please think about the information you might receive if you are in this situation in the future.

How much do you agree with the following statements? [RANDOM ORDER]

rec_most_fl: I receive pretty much all of the information that I need for my location.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

rec_miss_fl: Sometimes I miss information that I need for my location.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

rec_time: I receive new information about my location as soon as it is available.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

----------End Web pg ----------

[SHOW IF **rec_most_fl** < 3]

rec_most_exp: It sounds like you are not sure that you receive all the information you need when a flood threatens your location. Can you tell us what information you need but do not receive? [VERBATIM]

----------End Web pg ----------

rec_dif_sit: Can you think of a situation that might cause you to miss important forecast information if a flood were threatening your location? [VERBATIM]

----------End Web pg ----------

flood_hist: Have you ever experienced a flooding event?
0 - No

1 - Yes

----------End Web pg ----------

[SHOW IF **flood_hist** = 1]

Think about the most recent flooding event that you experienced.

flood_long: How long ago was it?
1 - Less than a year ago
2 - 1-2 years ago
3 - 2-5 years ago
4 - More than 5 years ago

flood_type: Do you remember what caused the flooding event?
1 - River flood - flooding caused by a river overflowing its banks
2 - Flash flood - sudden, intense flooding caused by heavy rainfall in a short time period
3 - Coastal flood - flooding caused by ocean water invading land areas along the coast due to storm surge, high tides, etc.
4 - Urban flood - flooding caused by overwhelmed drainage systems in urban areas
5 - Groundwater flood - flooding caused by saturated groundwater rising to the surface
6 - Dam/levee break flood - flooding caused by a dam or levee failure
7 - Don't know/don't remember

flood_act: Did you take some kind of protective action in response to the flooding event?
0 - No
1 - Yes

----------End Web pg ----------

[SHOW IF **flood_act** = 1]

flood_act_kind: What kind of protective action did you take in response to the flooding event? [VERBATIM]

flood_act_satis: Looking back, are you satisfied with the actions you took?
1 - Definitely no
2 - Probably no
3 - Not sure
4 - Probably yes
5 - Definitely yes

flood_act_again: How likely is it that you would take the same action again if you found yourself in the same situation in the future?
1 - Very unlikely
2 - Somewhat unlikely
3 - About as likely as not
4 - Somewhat likely
5 - Very likely

----------End Web pg ----------

[SHOW IF **flood_act_satis** < 3]

flood_act_exp: Can you briefly explain why you were not satisfied with the actions you took? [VERBATIM]

----------End Web pg ----------

People take many different actions in response to flooding events (e.g., look for more information, charge phones, prepare home, evacuate). Think about the actions you take when you get information that a flooding event may impact your location. If you have not been in this situation, please think about the actions you might take if you are in this situation in the future.

How much do you agree with the following statements? [RANDOM ORDER]

resp_ignore_fl: Sometimes I see information that says I could be at risk and decide to take my chances.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

resp_always_fl: I almost always take the protective actions that officials suggest, even if the actions are difficult (e.g., costly or time consuming).
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

resp_know_fl: Sometimes I am not sure what to do or what protective actions to take.
1 - Strongly disagree
2 - Disagree
3 - Neither disagree nor agree
4 - Agree
5 - Strongly agree

----------End Web pg ----------

resp_dif_sit: Can you think of a situation that might cause you to not take protective action if a flooding event were threatening your location? [VERBATIM]

----------End Web pg----------

[RANDOMIZE; 50% **home_ff_do** | 50% **car_ff_do**]

home_ff_do: Imagine that it is Saturday night at 8:00 PM and you are at home watching a movie. Suddenly, you see a FLASH FLOOD WARNING indicating that there is a high risk for flash flooding in the next 60 minutes. What would you do? Please be as specific as possible. [VERBATIM]

----------End Web pg----------

car_ff_do: Imagine that it is Saturday night at 8:00 PM and you are driving to the grocery store. Suddenly, you see a FLASH FLOOD WARNING indicating that there is a high risk for flash flooding in the next 60 minutes. What would you do? Please be as specific as possible. [VERBATIM]

----------End Web pg----------

ff_do_conff: How confident are you that your actions would protect you from the flash flood?
1 - Not at all confident
2 - Not very confident
3 - Somewhat confident
4 - Very confident

5 - Extremely confident

----------End Web pg----------

When flooding events threaten your area, how much do you rely on the following **sources** of information? If you have not been in this situation, how much would you plan to rely on the following **sources** of information if you were traveling in an area that is threatened by a flooding event? [RANDOM ORDER IN TABLE]

rely_nws: National Weather Service
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_loctv: Local TV stations
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_fam: Family, friends, neighbors, employers, co-workers, etc.
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_wx_chan: National TV stations like The Weather Channel or WeatherNation
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_em: State or local emergency managers
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

----------End Web pg----------

[SHOW IF **rely_nws** > 2]

You indicated that you rely on the National Weather Service for information about flooding events. How do you usually get information from the National Weather Service? Please select all that apply. [RANDOM ORDER CHECK BOXES; 1 = SELECTED]
nws_twit: X (formerly known as Twitter)
nws_face: Facebook
nws_web: Internet website
nws_tv: Television
nws_rad: Weather radio (NOAA/NWS radio)
nws_phone: Cell phone applications or notifications

nws_othr: Other (please specify)
nws_othr_spec: [VERBATIM]

----------End Web pg----------

[SHOW IF **rely_nws** = 1 OR 2]

rely_nws_why: You indicated that you do not rely on the National Weather Service for information about flooding events. Can you tell us why? [RANDOM ORDER]
1 - I don't know where to find the information
2 - The information is usually too complicated for me to understand
3 - They don't provide the information I want or need
4 - I don't trust the National Weather Service
5 - Other (please specify)
rely_nws_why_spec: [VERBATIM]

----------End Web pg----------

When flooding events threaten your area, how much do you rely on the following **channels** of information? Here, we define **channels** as tools or avenues of information. If you have not been in this situation, how much would you plan to rely on the following **channels** of information if you were traveling in an area that is threatened by a flooding event? [RANDOM ORDER IN TABLE]

rely_bdrad: Broadcast radio
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_wxrad: Weather radio (NOAA/NWS radio)
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_tv: Television
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_int: Internet websites focused on weather forecasts, such as those provided by the National Weather Service
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_soc: Social media, such as Facebook
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much

5 - Extensively

rely_wom: Word-of-mouth (including telephone calls or texts) from family, friends, neighbors, employers, co-workers, etc.
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

rely_phone: Automated text or phone notifications
1 - Not at all
2 - Not very much
3 - Somewhat
4 - Very much
5 - Extensively

----------End Web pg ----------

rely_tie: Please indicate which channel you rely on the most for information about flooding events.
1 - Broadcast radio
2 - Weather radio (NOAA/NWS radio)
3 - Television
4 - Internet websites focused on weather forecasts, such as those provided by the National Weather Service
5 - Social media, such as Facebook
6 - Word-of-mouth (including telephone calls or texts) from family, friends, neighbors, employers, co-workers, etc.
7 - Automated text or phone notifications

----------End Web pg ----------

Forecasters, websites, and phone applications often use maps to display information about flood risks.

flood_map_und: How would you rate your understanding of maps?
1 - Poor
2 - Fair
3 - Good
4 - Very good
5 - Excellent

flood_map_pref: Do you think maps are useful for displaying information about flood risks?
0 - No
1 - Yes

----------End Web pg----------

In addition to maps, forecasters often use probabilities to display information about flood risks.

For some people, probabilities can be difficult to calculate and interpret when you are trying to sort through a lot of information at once.

your_ability: How would you rate your ability to calculate and interpret probabilities when making decisions?
1 - Poor
2 - Fair
3 - Good
4 - Very good
5 - Excellent

----------End Web pg----------

[RANDOMIZE: **rand_timeline**: flooding event | flash flooding event]

Some people look for different kinds of information at different points in time. In the next few questions, we are going to give you a timeline and ask you to indicate the type of information that is *most* important at each point in time.

five_days: Five days before the **rand_timeline**
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

three_days: Three days before the **rand_timeline**
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

two_days: Two days before the **rand_timeline**
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

one_day: One day before the **rand_timeline**
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

zero_day: The day of the **rand_timeline**
1 - Location: what area is the flooding going to affect?
2 - Timing: when is the flooding going to happen?
3 - Chance: how likely is the flooding to occur?
4 - Severity: how much flooding is expected?
5 - Impacts: how might the flooding impact you and surrounding areas?
6 - Protective actions: how can you stay safe during the flood?

----------End Web pg----------

The NWS Weather Prediction Center (WPC) issues Excessive Rainfall Outlooks that indicate flash flooding threats across the contiguous United States. These Outlooks use a combination of words, colors, and numeric probabilities to represent threats. Here is one example:

[SHOW EXAMPLE OF ERO]

ero_see: Have you ever seen an Excessive Rainfall Outlook before?
0 - No
1 - Yes
2 - Not sure

ero_int_sub: How confident are you in your ability to understand the information presented in this Outlook?
1 - Not at all confident
2 - Slightly confident
3 - Moderately confident
4 - Very confident
5 - Extremely confident

ero_int_obj: Based on this Outlook, would you say that **location A** is…
1 - Very *unlikely* to experience flash flooding
2 - Somewhat *unlikely* to experience flash flooding
3 - Somewhat *likely* to experience flash flooding
4 - Very *likely* to experience flash flooding
5 - Not sure

ero_use: How likely are you to use an Outlook like this for making future decisions?
1 - Very unlikely
2 - Somewhat unlikely
3 - Unsure
4 - Somewhat likely
5 - Very likely

----------End Web pg ----------

[If **ero_int_sub** < 3]

[SHOW EXAMPLE OF ERO]

ero_int_no_why: You indicated that you are not confident in your ability to understand this Outlook. Can you tell us what aspects of the graphic are most confusing? [VERBATIM]

----------End Web pg ----------

[If **ero_use** < 3]

[SHOW EXAMPLE OF ERO]

ero_use_no: You indicated that you are unlikely or not very likely to use an Outlook like this for making future decisions. Can you tell us why? Please select all that apply.
ero_use_no_fnd: I don't know where to find forecasts like this.
ero_use_no_inacc: I believe forecasts like this are too inaccurate or unreliable.
ero_use_no_rel: Forecasts like this are not relevant to my daily life or activities.
ero_use_no_int: I find it hard to interpret forecasts like this.
ero_use_no_oth: Other (please specify)
ero_use_oth_spec: [verbatim]

----------End Web pg ----------

Now we are going to show you an example message from the National Weather Service and ask you how you might respond if you got this message. We know this can be difficult without more information. Just give us your first impression.

wea_rand_img: [RANDOMIZE: warning, emergency, warning_base, warning_cons, warning_cat]

If you saw this message and you were in the impacted area…

ibw_cncrn: How concerned would you be about the flash flood?
1 - Not at all concerned
2 - Not very concerned
3 - Somewhat concerned
4 - Very concerned
5 - Extremely concerned

ibw_act: Would you take actions to protect yourself from the flash flood?
1 - Definitely no
2 - Probably no
3 - Not sure
4 - Probably yes
5 - Definitely yes

ibw_act_conf: How confident are you that the actions you might take to protect yourself from the flash flood would be effective?
1 - Not at all confident
2 - Not very confident
3 - Somewhat confident
4 - Very confident
5 - Extremely confident

----------End Web pg ----------

Do you currently have any of the following types of insurance on your home? [Please indicate all that apply] [RANDOMIZED ORDER EXCEPT **ins_none** ALWAYS LAST]
ins_home: Standard homeowners insurance
ins_flood: Flood insurance
ins_none: None of the above/Not applicable

----------End Web pg----------

[SHOW ONLY IF **ins_home** = 1 AND **ins_flood** = 0]

home_cover: Does your homeowners insurance cover damage from flooding?
0 - No
1 - Yes
2 -Not sure

----------End Web pg ----------

[SHOW ONLY IF **ins_flood** = 0]

ins_flood_ever: Have you ever had flood insurance on your home?
0 - No
1 - Yes
2 - Not sure

----------End Web pg ----------

[SHOW ONLY IF **ins_flood** = 1]

flood_mortgage_ins: Are you required to have flood insurance as part of a mortgage (or escrow) on your home?
0 - No
1 - Yes
2 - Not sure

----------End Web pg ----------

As you may know, the Federal Emergency Management Agency (FEMA) is responsible for developing and maintaining a map of areas that are susceptible to flooding in the U.S. These areas are called Special Flood Hazard Areas or flood zones, or floodplains for short.

fema_flood_zone: To the best of your knowledge, is your home located in a FEMA flood zone?
0 - No
1 - Yes
2 - Not sure

----------End Web pg ----------

flood_damage: Has your home ever been damaged by a flooding event?
0 - No
1 - Yes
2- Not sure

----------End Web pg ----------

[SHOW ONLY IF **flood_damage** = 1]

flood_damage_yrs: Approximately how long ago did damage caused by the most recent flooding event occur?
1 - 0 - 5 years ago
2 - 6 -10 years ago
3 - 11-15 years ago
4 - 16 - 20 years ago
5 - 21 - 25 years ago
6 - More than 25 years ago

----------End Web pg ----------

flood_prob_1yr: If you had to guess, what is the probability that your home will be damaged by a flooding event next year?
1 - 0% (no chance)
2 - 1-10% (very low chance)
3 - 11-25% (low chance)
4 - 26-50% (medium chance)
5 - 51-75% (high chance)
6 - 76-100% (very high chance)

flood_prob_30yr: What is the probability that your home will be damaged by a flooding event in the next 30 years? Please indicate the probability as a percent between 0 and 100.
1 - 0% (no chance)
2 - 1-10% (very low chance)
3 - 11-25% (low chance)
4 - 26-50% (medium chance)
5 - 51-75% (high chance)
6 - 76-100% (very high chance)

----------End Web pg ----------

income: Was the estimated annual income for your household in 2023:
1 - Less than $50,000 [go to inc50]
2 - At least $50,000 but less than $100,000 [go to inc100]
3 - At least $100,000 but less than $150,000 [go to inc150]
4 - $150,000 or more [go to inc200]

----------End Web pg ----------

inc_50: Was the estimated annual income for your household in 2023:
1 - Less than $10,000
2 - $10,000 to less than $20,000
3 - $20,000 to less than $30,000
4 - $30,000 to less than $40,000
5 - $40,000 to less than $50,000

----------End Web pg ----------

inc_100: Was the estimated annual income for your household in 2023:
6 - $50,000 to less than $60,000
7 - $60,000 to less than $70,000
8 - $70,000 to less than $80,000
9 - $80,000 to less than $90,000
10 - $90,000 to less than $100,000

----------End Web pg ----------

inc_150: Was the estimated annual income for your household in 2023:
11 - $100,000 to less than $110,000
12 - $110,000 to less than $120,000
13 - $120,000 to less than $130,000
14 - $130,000 to less than $140,000
15 - $140,000 to less than $150,000

----------End Web pg ----------

inc_200: Was the estimated annual income for your household in 2023:
16 - $150,000 to less than $160,000
17 - $160,000 to less than $170,000
18 - $170,000 to less than $180,000
19 - $180,000 to less than $190,000
20 - $190,000 to less than $200,000
21 - $200,000 or more

----------End Web pg ----------

edu: What is the highest level of education you have COMPLETED?
1 - Less than high school
2 - High school / GED
3 - Vocational or Technical Training
4 - Some College; NO degree
5 - 2-year College / Associate's degree
6 - Bachelor's Degree
7 - Master's Degree

8 - PhD / JD (Law) / MD

----------End Web pg ----------

comments: Is there anything else that you would like us to know about how you receive, understand, or respond to information about flooding?